GENERAL PRINCIPLES TO CALCULATE FINANCED EMISSIONS FOR NET ZERO PORTFOLIO ALIGNMENT

Guidelines

Santander Group

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1 INTRODUCTION

1.1 Purposes

These guidelines explain the general principles to be applied when calculating financed emissions for portfolio net zero alignment purposes.

1.2 Definition

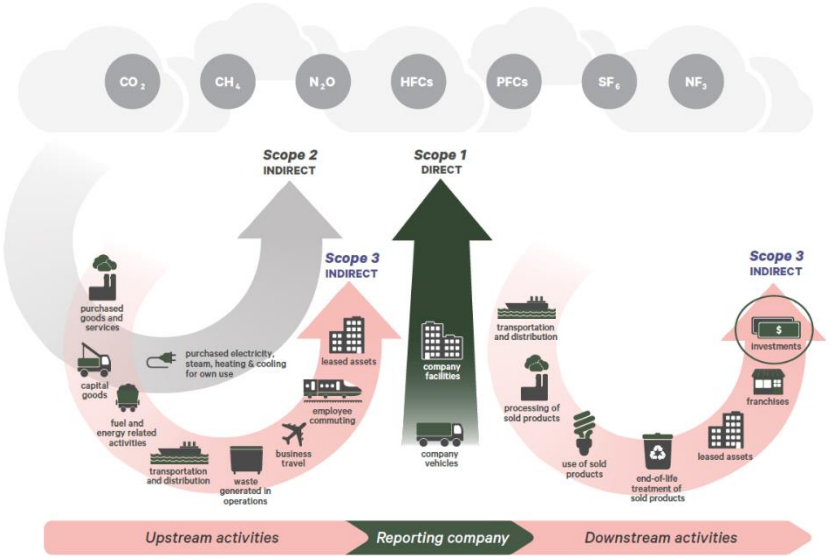
Measuring financed emissions is the process of emissions measurement from the activities/customers we finance/invest in. This process of emissions measurement could be used as part of the portfolio alignment to set decarbonization targets or just for disclosure purposes. In these guidelines, we are focusing on the first use.

For a proper understanding of these guidelines, the following terms/concepts should also be defined:

- Net zero: climate objective of reaching net zero carbon or Greenhouse Gas (GHG) emissions by a certain date, normally 2050, committed to by many corporates and organizations. Considered by many scientists as the needed goal to limit the temperature warming to 1.5º degrees Celsius by the end of the century, compared to pre-industrial levels. Banco Santander announced in February 2021 its ambition to achieve net zero carbon emissions across the group by 2050. The ambition applies to the group's own operations, which have been carbon neutral since 2020, and all client emissions that result from any lending, advisory or investment services provided by Santander.
- Greenhouse gas emissions (GHG): the seven gases mandated under the Kyoto Protocol and are to be included in national inventories under the United Nations Framework Convention on Climate Change (UNFCCC): carbon dioxide (CO_2), methane (CH_4), nitrous oxide (N_2O), hydrofluorocarbons (HFCs), perfluorocarbons (PFCs), sulfur hexafluoride (SF_6) and nitrogen trifluoride (NF_3). For ease of accounting, these gases are usually converted to and expressed as carbon dioxide equivalents (CO_2e). The internationally recognised standard to measure GHG emissions is the GHG Protocol[1] and it defines the following scopes of emissions.
    - Scope 1: Direct GHG emissions that occur from sources owned or controlled by the reporting company, i.e., emissions from combustion in owned or controlled boilers, furnaces, vehicles, etc.
    - Scope 2: Indirect GHG emissions from the generation of purchased or acquired electricity, steam, heating, or cooling consumed by the reporting company. Scope 2 emissions physically occur at the facility where the electricity, steam, heating, or cooling is utilized.
    - Scope 3: All other indirect GHG emissions (not included in Scope 2) that occur in the value chain of the reporting company. Scope 3 can be broken down into upstream emissions and downstream emissions. Upstream emissions include all emissions that occur in the life cycle of a material/product/service up to the point of sale by the producer, such as from the production or extraction of purchased materials. Downstream emissions include all emissions that occur as a

[1] See the "The GHG Protocol Corporate Accounting and Reporting Standard" (https://ghgprotocol.org/corporate-standard)

consequence of the distribution, storage, use, and end-of-life treatment of the organization's products or services.



Source: (WRI and WBCSD, 2011)

- Financed emissions: measurement of the emissions coming from our customers' activities weighted by the financing we grant them. Financed emissions might be calculated for lending and investment activities, although we are prioritising the financed emissions coming from our lending activities, as the ones more material for Santander. Financed emissions are the main way to report emissions from the "Scope 3: Category 15 – Investments"[2], which are considered to be the most material source of GHG emissions for the bank. Santander's Scope 3: Category 15 emissions are the attributed emissions of our lending/investment customers' activities. Therefore, Santander's financed emissions are the attributed Scope 1, 2 and 3 from our customers. In many cases to obtain estimations of Scope 3 emissions from our customers is very challenging and considered only necessary for sectors where most of the emissions occur in their value chain (for example: oil and gas, coal mining, in the use of their products, i.e. burning fuels…). As a result, financed emissions calculations should include a rationale detailing the scope of the client emissions included.
- PCAF[3]: a private "industry-led partnership to facilitate transparency and accountability of the financial industry to the Paris Agreement", made of financial institutions that work together to develop and implement a harmonised approach to calculate and disclose the GHG emissions associated with their loans and investments. The standard provides different ways to measure financed emissions, depending on the sector/type of asset under analysis, and on the availability of information and also providing a metric to calculate the level of quality of the resulting emissions calculation. Created in 2015, Banco Santander joined PCAF in September

[2] See the "The Corporate Value Chain (Scope 3) Accounting and Reporting Standard", from GHG Protocol. (https://ghgprotocol.org/corporate-value-chain-scope-3-standard)
[3] Partnership for Carbon Accounting Financials (https://carbonaccountingfinancials.com)

2021, and therefore committed to use their standard in measuring and disclosing financed emissions. PCAF offers its members access to a database with several emissions factors to start calculating financed emissions in the absence of higher quality and more customer-specific emissions data. Sometimes it is said informally that in some situations we "use PCAF" meaning we are calculating financed emissions with this database, although in almost all the calculations we are following the PCAF standard, as the main reference.

- Portfolio alignment: is one of the four pillars of Santander's climate strategy. It seeks to align our portfolios to meet the Paris Agreement goals, and therefore to contributing to limit the temperature increases to 1.5º C by reaching net zero emissions by 2050. Santander's portfolio alignment strategy can be split into 4 different stages, i) assess emissions/baseline (the scope of this document); ii) set targets and pathways; iii) engagement with customers and/or on public policy; and iv) steer portfolio through the mobilisation of internal and external levers. Consistent with broad stakeholders' expectations and the Net Zero Banking Alliance (NZBA), portfolio alignment objectives materialise in sectoral decarbonization strategies and 2030 net zero decarbonization targets.
- Net Zero Banking Alliance (NZBA[4] or the Alliance): the United Nations convened alliance of banks, currently representing over 40% of global banking assets, which are committed to aligning their lending and investment portfolios with net-zero emissions by 2050. NZBA was founded in April 2021. Combining near-term action with accountability, this ambitious commitment sees signatory banks setting intermediate targets for 2030 or sooner using robust, science-based guidelines. The Alliance also executes coordinated advocacy and alignment as the banking element of the Glasgow Financial Alliance for Net Zero and the climate-focused element of the Principles for Responsible Banking.
- Facilitated emissions (Capital Markets): these are the emissions from capital markets activities (i.e. advisory services of the bank). PCAF is developing a standard with guidance on how to assess facilitated emissions, and as contemplated by the NZBA founding papers, it will be included in the NZBA guidelines. The Capital Markets' PCAF standard will focus on the facilitation activity of the book runners and managers in a capital markets issuance.
    - Facilitator: An institution (usually large international banks) that helps an entity (in this paper, generally corporates) to issue equity or debt instruments in the capital markets. The facilitator may carry out activities including advising the issuing entity on structure, pricing, and process; preparing materials for and engaging with investors; and arranging and guiding issuing entities on a roadshow. Formal roles encompassed by this term include Lead/Active/Passive Bookrunner(s) and Lead/Co-Manager(s).

1.3 Scope

These guidelines are prepared by Banco Santander, S.A., in its condition of parent company of Santander Group, resulting directly applicable to the Corporate Center, and is provided to the entities comprising the Group as a reference document, establishing the principles to be followed to the subject to which it refers.

[4] https://www.unepfi.org/net-zero-banking/

Financed emissions measurement may have different purposes and uses. Currently, the principal ones in the banking industry can be divided between “portfolio alignment” and “reporting/disclosure” purposes. This document focuses on portfolio alignment and disclosure purposes.

Currently, there is a regulatory requirement for the Group to report financed emissions for a broad scope of lending activities. For this “reporting/disclosure” purposes, there are specific internal guidelines to follow “Pilar 3 ESG - Functional guidelines”. This is relatively consistent with this one, although offers solutions to address data gaps and calculate financed emissions with less accuracy, compared with what we recommend for portfolio alignment.

While Pillar 3 Reporting uses emissions data on sector level and Portfolio Alignment requires more granularity on asset level, both methodologies for measuring financed emissions should converge over time to ensure synergies and cost efficiencies, always leveraging the most advanced methodology and data across both initiatives.

These guidelines set general principles to follow for lending portfolios, with recommendations on what choices should be made when defining the specific methodology to execute for a given portfolio/sector.

These guidelines should be complemented with other implementation guidelines that define this specific methodology for each portfolio. The business units or functions conducting financed emissions calculations should prepare implementation guidelines to explain the details of the exercise execution covering aspects like the methodology followed, data used, systems, key assumptions, etc.

Although it would be desirable to have financed emissions measurement more frequently than yearly, it is difficult to obtain emissions data from corporates more frequently than for the year-end reporting. Therefore, the frequency of financed emissions, when applicable, should be annually at a minimum.

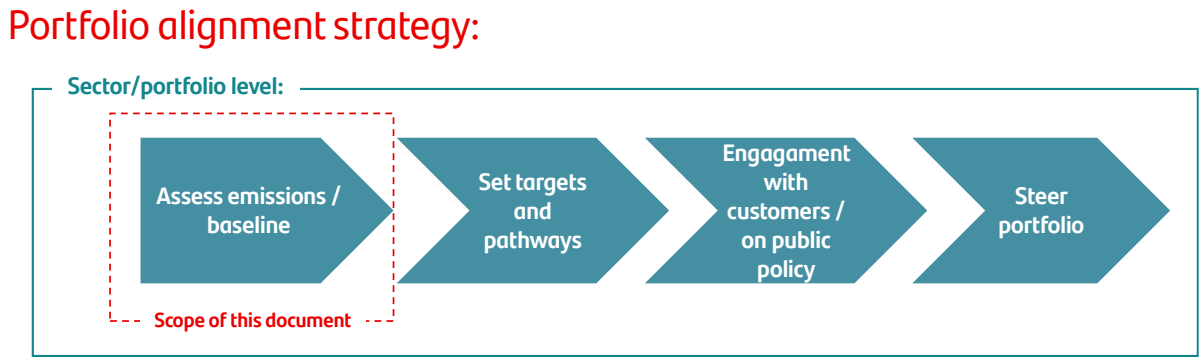
2 GENERAL PRINCIPLES

The aim of this document is to set general principles to follow when calculating financed emissions for portfolio alignment and disclosure purposes.

We use an internal methodology for the calculation of the financed emissions, following the PCAF standard ("The Global GHG Accounting and Reporting Standard for the Financial Industry") as a guiding principle for our methodology to calculate financed emissions.

This is not the only approach, some banks have developed methodologies to calculate financed emissions based on PACTA (Paris Agreement Capital Transition Assessment, from 2Dii[5]), future guidelines reviews should assess developments in this field and whether additional approaches could also be considered.

This is the general process or portfolio alignment's strategy:



When calculating financed emissions for portfolio alignment, it is key to first define the scope of the measurement in terms of emissions' scopes, sector, parts of the value chain, countries' exposure, segments, products, and others.

2.1 Global and local sectors

Considering NZBA current requirements, which Santander (and other banks) committed to for net zero alignment, we identify two different groups of sectors to address for financed emissions measurement and portfolio alignment:

- Global sectors: The companies from these sectors are generally multinational corporates that operate in several countries or have operations in strategic sectors. The more carbon-intensive sectors, measured in financial or physical intensity metrics, are usually global sectors. Most of Santander's exposure to global sectors is within the SCIB segment, and therefore SCIB leads this work globally.
- Local sectors: Customers and companies that are mainly based in a single country, and in many cases the emissions come from assets located in a single country. Most of Santander's exposure to local sectors are within commercial banking segments (individuals, consumer finance, small-

[5] https://2degrees-investing.org/resource/pacta/

and medium-sized enterprises - SMEs, or corporates). The work in these sectors is split into sectors and geographies, and Local Business leads are normally in charge of portfolio alignment based on financed emissions data measured and delivered by corresponding support areas.

2.2 The different asset classes

The PCAF's standard comprises seven different asset classes to measure financed emissions, although for the current scope of work we only consider five of them, as per PCAF's definitions:

- Business loans: include all on-balance sheet loans and lines of credit to businesses, non-profits, and any other structure of organization that are not traded on a market and are for general corporate purposes, i.e., with unknown use of proceeds as defined by the GHG Protocol.
- Project finance: includes all on-balance sheet loans or equities to projects or activities that are designated for specific purposes, i.e., with known use of proceeds as defined by the GHG Protocol. The financing is designated for a defined activity or set of activities, such as the construction and operation of a gas-fired power plant, a wind or solar project, or energy efficiency projects. To calculate emissions, only the financed (ring-fenced) activities are included. Emissions and financials related to existing activities outside the financed project but within the financed organization are not considered.
- Commercial real estate: includes on-balance sheet loans for specific corporate purposes, namely the purchase and refinance of commercial real estate (CRE), and on-balance sheet investments in CRE when the financial institution has no operational control over the property. This definition implies that the property is used for commercial purposes, such as retail, hotels, office space, industrial, or large multifamily rentals. In all cases, the owner of the building uses the property to conduct income-generating activities. (Construction is considered out of the scope of CRE activities)
- Mortgages: includes on-balance sheet loans for specific consumer purposes—namely the purchase and refinance of residential property, including individual homes and multifamily housing with a small number of units. This definition implies that the property is used only for residential purposes and not for commercial activities.
- Motor vehicle loans: refers to on-balance sheet loans and lines of credit to businesses and consumers for specific (corporate or consumer) purposes - namely the finance of one or several motor vehicles.

2.3 Attribution

As per PCAF's standard: "According to the GHG Protocol Corporate Value Chain (Scope 3) Accounting and Reporting Standard, GHG emissions from loans and investments (Category 15) should be allocated to the reporting financial institutions based on the proportional share of lending or investment in the borrower or investee. Attribution is based on the annual emissions of the borrower and investee; as a result, GHG emissions are reported on at least an annual basis.

The methodologies in the (PCAF) Financed Emissions Standard apply the same general attribution principles across all asset classes:

1. Financed emissions are always calculated by multiplying an attribution factor (specific to that asset class) by the emissions of the borrower or investee.

2. The financed emissions are defined as the share of total annual GHG emissions of the borrower or investee that is allocated to the loan(s) or investment(s).
3. The attribution factor is calculated by determining the share of the outstanding amount of loans and investments of a financial institution over the total equity and debt of the company, project, etc. to which the financial institution has lent money or in which it has invested capital."

Example for the business loans asset class:

$$\text{Financed emissions} = \sum_i \text{Attribution factor}_i \times \text{Emissions}_i \quad (\text{with } i = \text{borrower or investee})$$

$$\text{Attribution factor}_i \rightarrow \frac{\text{Outstanding amount}_i}{\text{Total equity + debt}_i}$$

2.4 Data quality

Data quality and availability to measure financed emissions are two very important challenges to address. Given that measuring financed emissions is a nascent field, it is common to use information with some time lag or that might lack accuracy.

PCAF's standard has a scale to measure data quality that goes from 1 (the best) to 5 (the worst), applicable to each one of the asset classes included within the standard.

PCAF score of 1 or 2 are actual or reported emissions from the company or asset. This high quality data is desirable to know what the real emissions from our customers are, and allow us to monitor decarbonization progress of our customers, for portfolio alignment purposes.

PCAF score of 4 or 5 are emissions estimated from sectoral averages based on very limited available information. These attributed emissions are estimated from some available information from the customers/assets. In these circumstances, the bank cannot know real emissions or decarbonization progress over time.

PCAF score of 3 may capture some of the customer or asset specific climate data that may help the bank understand the level of emissions and see some progress overtime. Although there should be a plan to improve quality of data, aiming to reach PCAF score of 1 or 2, if plausible.

PCAF score of 3/4 should be the minimum threshold to use the financed emissions for portfolio alignment. To be able to monitor real progress of our customers' decarbonization. Although, for some asset classes it is not possible to reach a PCAF score of 1 or 2 in the current situation, and the PCAF score could go beyond 4.

2.5 Methodology

2.5.1 Business loans

The financed emissions from business loans and unlisted equity are calculated by multiplying the attribution factor by the emissions of the borrower or investee company and then taking the sum of these emissions.

In many cases, the financed emissions are calculated at the customers' holding or group level rather than at the subsidiary level for groups, given that reported emissions or production are usually available on a consolidated basis.

The attribution factor represents the proportional share of a given company—that is, the ratio of the outstanding amount to total equity and debt for private companies and Enterprise Value Including Cash (EVIC) for listed companies.

$$\text{Counterparty's financed emissions} = \text{Counterparty's emissions} * \frac{\text{Counterparty's outstanding}}{\text{Counterparty's equity} + \text{Counterparty's debt}}$$

Santander guidance: within the business loans asset class, the PCAF standard recommends using as denominator the "EVIC" for listed companies and "Total Equity + debt" or unlisted companies. Although we recommend using the latter for either listed or unlisted companies, as we think it avoids volatility year over year.

Using the PCAF methodology for financed emissions, there are several ways to calculate them, depending on the availability of counterparty's specific financial and emissions data.

Depending on the information used for the calculation of emissions, according to the PCAF standards, different data quality scores are defined from 1 -highest data quality- to 5 - lowest data quality-. The following table provides data quality scores for each of the described options and sub-options (if applicable) that can be used:

Data quality	**Options to estimate the financed emissions**		**When to use each option**
Score 1	**Option 1: Reported emissions**	1a	- Outstanding amount in the company and total company equity plus debt are known. - **Audited emissions of the company** are available
Score 2		1b	- Outstanding amount in the company and total company equity plus debt are known. - **Non audited emissions calculated by the company** are available.
	Option 2: Physical activity based emissions	2a	- Outstanding amount in the company and total company equity plus debt are known. - Reported company emissions are not known. - **Emissions are calculated** using primary physical activity data for the **company's energy consumption** and emission factors specific to that primary data.
Score 3		2b	- Outstanding amount in the company and total company equity plus debt are known. - Reported company emissions are not known.

Data quality	Options to estimate the financed emissions		When to use each option
			- **Emissions are calculated** using primary physical activity data for the **company's production** and emission factors specific to that primary data
Score 4	**Option 3: Economic activity based emissions**	3a	- Outstanding amount in the company, total company equity plus debt, and the company's revenue are known. - **Emission factors** for the sector **per unit of revenue are known** (e.g., tCO2e per euro of revenue earned in a sector).
Score 5		3b	- Outstanding amount in the company is known. - **Emission factors** for the sector **per unit of asset** (e.g., tCO2e per euro of asset in a sector) are known.

Santander guidance: for the purposes of portfolio alignment, option 1 – reported emissions should be used. The rationale for option 1 is that other options only allow us to have a rough estimate of counterparties' emissions, and it wouldn't be possible to monitor true customer decarbonization progress without more accurate estimation of their emissions. However, given that emissions measurement is evolving quickly, it is reasonable to start working with lower-quality PCAF scores, aiming to improve the quality as quickly as possible.

2.5.2 Project finance

The financed emissions from a single project are calculated by multiplying the attribution factor by the emissions of the respective project. The total financed emissions from multiple projects are calculated using the following equation:

$$\text{Project financed emissions} = \text{Project's reported/estimated emissions} * \frac{\text{Outstanding amount}}{\text{Total equity + Debt}}$$

Santander guidance: the PCAF standard recommends using as the denominator the "equity + debt" of the project finance. Projects are usually structured in special purpose vehicle (SPV), which is a legal entity managed or related to other bigger corporations. SPV entities typically do not produce regular financial statements as big corporations do. As a result, calculation may need to rely on equity and debt figures provided at the deal sign-off. This is acceptable if the figures allow making a reasonable estimation of the attribution factor.

Data quality	Options to estimate the financed emissions		When to use each option
Score 1	**Option 1: Reported emissions**	**1a**	- Outstanding amount in the project and total project equity plus debt are known. - **Verified emissions of the project** are available.

Data quality	Options to estimate the financed emissions		When to use each option
Score 2		1b	- Outstanding amount in the project and total project equity plus debt are known. - **Unverified emissions reported** by the project are available.
	Option 2: Physical activity-based emissions	2a	- Outstanding amount in the project and total project equity plus debt are known. - Project emissions are not known but calculated using primary physical activity data for the **project's energy consumption** and emission factors specific to that primary data. - Relevant process emissions are added.
Score 3		2b	- Outstanding amount in the project and total project equity plus debt are known. - Project emissions are not known. - Emissions are calculated using primary physical activity data for the **project's production** and emission factors specific to that primary data.
Score 4	Option 3: Economic activity-based emissions	3a	- Outstanding amount in the project, total project equity plus debt, and the project's revenue are known. - **Emission factors** for the sector **per unit of revenue** or from similar projects is known (e.g., tCO2e per euro of revenue earned in a sector).
Score 5		3b	- Outstanding amount in the project is known. - **Emission factors** for the sector **per unit of asset** or economic activity-based emission factors from similar projects (e.g., tCO2e per euro of asset in a sector) are known.
		3c	- Outstanding amount in the project is known. - **Emission factors** for the sector **per unit of revenue** (e.g., tCO2e per euro of revenue earned in a sector) **and asset turnover ratios** for the sector or from similar projects are known.

Santander guidance: similar to the "business loans" asset class, for the purposes of portfolio alignment, option 1 – reported emissions should be used. Other options only allow for a rough estimate of the project's emissions. Since each project will normally have a single technology used in its activities, using option 2 (and the use of emissions factors) is reasonable if estimation is required. Since emissions measurement is evolving quickly, it is reasonable to start working with lower-quality PCAF scores, aiming to improve the quality as quickly as possible.

2.5.3 Commercial real estate

The financed emissions from of a CRE loan or investment are calculated by multiplying the attribution factor by the emissions of the building. Thus, financed emissions are calculated as follows:

$$\text{Building financed emissions } (tCO2e) = \text{Building emissions} * \frac{\text{Building loan outstanding}}{\text{Property value at origination}}$$

Santander guidance: the CRE sector is one of the most challenging sectors for financed emissions estimation for several reasons. One challenge is that sector boundaries choices must be made. Banks have prioritised the collaterals databases for risk management purposes and not so much for the specific activities carried by the customers with the collateralised assets. To establish the perimeter for emissions estimation, CRE assets must be further defined by type of activity such as "*for […] retail, hotels, office space, industrial, or large multifamily rentals*." In all cases, the owner of the building must use the property to conduct income-generating activities. Construction is considered out of the scope of CRE activities.

Data quality	Options to estimate the financed emissions		When to use each option
Score 1	**Option 1: Actual building emissions**	**1a**	- Primary data on **actual building energy consumption** (i.e., metered data) is available. - Emissions are calculated using actual building energy consumption and **supplier-specific emission factors** specific to the respective energy source.
Score 2		**1b**	- Primary data on **actual building energy consumption** (i.e., metered data) is available. - Emissions are calculated using actual building energy consumption and **average emission factors specific to the respective energy source.**
Score 3	**Option 2: Estimated building emissions based on floor area**	**2a**	- **Estimated building energy consumption per floor area based on official building energy labels and the floor area are available**. - Emissions are calculated using estimated building energy consumption and **average emission factors** specific to the respective energy source.
Score 4		**2b**	- **Estimated building energy consumption per floor area based on building type and location-specific statistical data and the floor area are available.** - Emissions are calculated using estimated building energy consumption and **average emission factors specific** to the respective energy source.
Score 5	**Option 3: Estimated building emissions based on number of buildings**	**3**	- **Estimated building energy consumption per building based on building type and location-specific statistical data and the number of buildings are available.** - Emissions are calculated using estimated building energy consumption and **average emission factors** specific to the respective energy source.

Santander guidance: for Real Estate assets there is a lack of and/or limited reliability/homogeneity of energy labels or energy performance certificates (EPC) in many geographies. We expect new regulations in the coming years to enforce the reliability/homogeneity of EPCs for real estate owned

assets. When estimating the financed emissions of a CRE portfolio it is common to have EPC data for only a part of the portfolio or not all, as well as to have outdated EPC information as it is not updated frequently. If EPC data is not available, models can be used to estimate energy efficiency based on the property location and other available data.

In regard to the individual inputs for the PCAF formula for the *property value at origination,* where not feasible or possible to obtain for certain properties, fallback values such as the property value at the latest valuation date or an equivalent valuation approach can be used*

** as per PCAF guidance: When the property value at loan or equity origination is not feasible to obtain, financial institutions shall use the latest property value available and fix this value for the following years of GHG accounting, i.e., the denominator remains constant from the first year of GHG accounting onward*

2.5.4 Mortgages

The financed emissions of mortgages are calculated by multiplying the attribution factor by the emissions of the building. Thus, financed emissions are calculated as follows:

$$\text{Building financed emissions } (tCO2e) = \text{Building emissions} * \frac{\text{Building loan outstanding}}{\text{Property value at origination}}$$

Data quality	Options to estimate the financed emissions		When to use each option
Score 1	**Option 1: Actual building emissions**	**1a**	- Primary data on **actual building energy consumption** (i.e., metered data) is available. - Emissions are calculated using actual building energy consumption and **supplier-specific emission factors** specific to the respective energy source.
Score 2		**1b**	- Primary data on **actual building energy consumption** (i.e., metered data) is available. - Emissions are calculated using actual building energy consumption and **average emission factors specific to the respective energy source.**
Score 3	**Option 2: Estimated building emissions based on floor area**	**2a**	- **Estimated building energy consumption per floor area based on official building energy labels and the floor area are available**. - Emissions are calculated using estimated building energy consumption and **average emission factors** specific to the respective energy source.
Score 4		**2b**	- **Estimated building energy consumption per floor area based on building type and location-specific statistical data and the floor area are available.**

Data quality	Options to estimate the financed emissions		When to use each option
			- Emissions are calculated using estimated building energy consumption and **average emission factors specific** to the respective energy source.
Score 5	**Option 3: Estimated building emissions based on number of buildings**	**3**	- **Estimated building energy consumption per building based on building type and location-specific statistical data and the number of buildings are available.** - Emissions are calculated using estimated building energy consumption and **average emission factors** specific to the respective energy source.

Santander guidance: for mortgage assets there is a lack of and/or limited reliability/homogeneity of energy labels or energy performance certificates (EPC) in many geographies. We expect new regulations in the coming years to enforce the reliability/homogeneity of EPCs for residential real estate owned assets. When estimating the financed emissions of a mortgage portfolio it is common to have EPC data for only a part of the portfolio or not all, as well as to have outdated EPC data as it is not updated frequently. If EPC data is unavailable, models can be used to estimate energy efficiency based on location and other available data.

In regard to the individual inputs for the PCAF formula, for *property value at origination*, where not feasible or possible to obtain for certain properties, fallback values such as the property value at the latest valuation date or an equivalent valuation approach can be used*

** as per PCAF guidance: When the property value at loan or equity origination is not feasible to obtain, financial institutions shall use the latest property value available and fix this value for the following years of GHG accounting, i.e., the denominator remains constant from the first year of GHG accounting onward.*

2.5.5 Motor vehicle loans

The financed emissions from a motor vehicle loan are calculated by multiplying the attribution factor by the emissions of the motor vehicle. The total financed emissions from multiple motor vehicle loans are calculated as follows:

$$\text{Vehicle financed emissions } (tCO2e) = \text{Vehicle emissions} * \frac{\text{Vehicle loan outstanding}}{\text{Total value at origination}}$$

Santander guidance: Santander's two most material businesses related to motor vehicles are: (1) lending auto manufacturers to build cars; and (2) personal auto loans (and leases) to finance the acquisition of a motor vehicle. For these two businesses, the sources of emissions and ways to calculate financed emissions are different:

- Personal auto loans: the “motor vehicle loans” asset class is used to calculate the financed emissions from the vehicles financed.
- Auto manufacturers: the “motor vehicle loans” asset class is used to calculate the financed emissions from the vehicles produced and sold by the auto manufacturers. This is Scope 3 – Category 11 “use of sold products” emissions. This is then treated in line with the global sectors as described in Sec. 2.1 of this document .

The material emissions of the vehicles are:

- Scope 1: direct emissions from fuel combustion in vehicles (All cars will have Scope 1 emissions, except electric and hydrogen cars, which emissions Scope 1 are 0)
- Scope 2: indirect emissions from electricity generation consumed in EVs (Only electric cars and plug-in hybrid cars will have Scope 2 emissions)

Emission factors are to include Scope 1 and Scope 2 emissions as per above. In some instances, and depending on vehicle and fuel type, the vehicle specific emissions provided by recognised data sources, such as the WLTP (Worldwide Harmonized Light Vehicles Test Procedures) and NEDC (New European Driving Cycle), ought to be supplemented by emissions data (such as central government’s emissions factors) to cover the respective Scope 1 and Scope 2 emissions.

Data quality	**Options to estimate the financed emissions**		**When to use each option**
Score 1	**Option 1: Actual vehicle-specific emissions**	**1a**	- Outstanding amount and total value at origination of vehicle or vehicle fleet are known. - Primary data on **actual vehicle fuel consumption is available**. - Emissions are calculated using actual fuel consumption and fuel type-specific emission factors.
		1b	- Outstanding amount and total value at origination of vehicle or vehicle fleet are known. - Vehicle efficiency and fuel type (fossil and/or electricity) are available from **known vehicle make and model**. - Primary data on **actual vehicle distance traveled is available**. - Emissions are calculated using estimated fuel consumption and fuel type-specific emission factors.
Score 2	**Option 2: Estimated vehicle-specific emissions**	**2a**	- Outstanding amount and total value at origination of vehicle or vehicle fleet are known. - Vehicle efficiency and fuel type (fossil and/or electricity) are available from **known vehicle make and model**. - **Distance traveled is estimated based on local statistical data.** - Emissions are calculated using estimated fuel consumption and fuel type-specific emission factors.
Score 3		**2b**	- Outstanding amount and total value at origination of vehicle or vehicle fleet are known. - Vehicle efficiency and fuel type (fossil and/or electricity) are available from **known vehicle make and model**. - **Distance traveled is estimated based on regional statistical data.**

Data quality	Options to estimate the financed emissions		When to use each option
			- Emissions are calculated using estimated fuel consumption and fuel type-specific emission factors.
Score 4	**Option 3: Estimated vehicle-unspecific emissions**	**3a**	- Outstanding amount and total value at origination of vehicle or vehicle fleet are known. - Vehicle efficiency and fuel type (fossil and/or electricity) are estimated from **known vehicle type (vehicle make and model are unknown).** **- Distance traveled is estimated based on local or regional statistical data**. - Emissions are calculated using estimated fuel consumption and fuel type-specific emission factors.
Score 5		**3b**	- Outstanding amount and total value at origination of vehicle or vehicle fleet are known. - Vehicle efficiency and fuel type (fossil and/or electricity) are estimated for an **average vehicle (vehicle make and model and vehicle type are unknown**). **- Distance traveled is estimated based on local or regional statistical data**. - Emissions are calculated using estimated fuel consumption and fuel type-specific emission factors.

2.5.6 Types of financed emissions metrics

There are three main types of financed emissions metrics:

- Absolute emissions: The total GHG emissions of an asset class or portfolio.
- Economic emission intensity: Absolute emissions divided by the loan or investment volume in EUR, expressed as tCO_2e/€M loaned or invested.
- Physical emission intensity: Absolute emissions divided by a value of physical activity or output, expressed as, e.g., tCO_2e/MWh, tCO_2e/tonne product produced.

To calculate the economic emission intensity of a portfolio or an asset, we need to calculate both the emissions and the physical activity or output (e.g. MW for Power generation, m^2 for mortgages, km for auto, etc…) for each component of the portfolio (customer, project or asset, generally), and then calculate economic emission intensity weighted by the physical activity or output.

Santander guidance: when a decarbonization target has been set for a portfolio, it is needed to report yearly the three types of emissions metrics.

2.5.7 Emissions scoping, parts of the value chain and asset class to use by sector

When calculating financed emissions it is important to extend the calculations to the emissions' scopes and parts of the value chain that are material and relevant for each sector. Considering the

materiality of the scopes' emissions for each sector, weighing the feasibility of calculations and where the decarbonization efforts could be put on.

Within a sectoral classification there can be high heterogeneity of activities and emissions-intensity. This is particularly relevant in the global sectors. To address this, many banks like Santander are calculating the financed emissions of the most polluting or relevant parts of each sector.

There are some difficult sectors that could be addressed, for emissions measurement, with more than one asset class (as defined in the previous section).

The Santander guidance for scoping and asset classes to use are:

SECTOR	Activity within the value chain	Emissions' scopes	Asset class
Coal mining	Thermal coal mining	Scope 1, 2 and 3 (use of sold products)	Business loans / Project finance
Power generation	Generation	Scope 1	
Energy (Oil&Gas)	Upstream/generation, and integrated with upstream activities	Scope 1, 2 and 3 (use of sold products)	
Steel	Manufacturing	Scope 1 and 2	
Aluminium	Manufacturing	Scope 1 and 2	
Cement	Manufacturing	Scope 1 and 2	
Aviation	Airliners	Scope 1 and 2	
Shipping	Maritime transport	Scope 1 and 2	
Auto	Auto manufacturing	Scope 3 (use of sold products)	Business loans / Motor vehicle loans
	Final consumers (personal auto-loans)	Scope 1 and 2	Motor vehicle loans
Commercial Real Estate	Income generating properties	Scope 1 and 2	Commercial real estate
Agriculture	Primary production of Agricultural and livestock activities	Scope 1 and 2*	Business loans (and individuals loans)
Mortgages	Residential properties	Scope 1 and 2	Mortgages

*: the agriculture sector is one of the most complex to calculate the financed emissions. There are no current methodologies in place for banks but there are some initiatives and few banks have disclosed

decarbonization targets for this sector. “Land use change” could be a very material source of emissions for this sector, and to measure this is important to capture history of forest-to-non-forest conversions in financed farms, and also to potentially include removals in some cases (e.g. forest regeneration, well-managed pasture, etc.). Although removals accounting for target setting is under high scrutiny and should be reported separately.

2.6 Other standards and guidance

There are several external and internal standards and guidelines that can complement the content of these guidelines. Emissions measurement is evolving quickly, and we will regularly update these guidelines to include new reference standards and guidance that might be published in the near future.

- **PCAF standard**: is the main external reference to calculate the financed emissions. Santander is a member of PCAF and aims to follow the PCAF standard. This document is based on the PCAF standard.
- **NZBA sectoral white papers**: the Alliance is conducting sectoral working groups for the sectors included in the “CCCA Guidelines for Climate Change Target Setting”. In these white papers there are recommendations (not compulsory and not prescriptive) on methodological choices to be made in terms of sector boundaries, emissions scoping, etc., when setting decarbonization targets for these sectors.
- **SCIB Portfolio alignment implementation notes**: these internal guidelines set specific implementation criteria to calculate the financed emissions to set decarbonization targets for SCIB portfolios. The SCIB guidelines are consistent and comply with the general criteria explained in this document.
- **Pilar 3 ESG - Functional guidelines**: these internal guidelines set also implementation criteria but with a different scope and use. The purpose is regulatory disclosure (Pillar 3) and with a broader scope as it requests to cover all non-financial corporations’ exposure. These regulatory requirements have a phase-in approach until June-2024.
- ***Other potential implementation guidelines***: Business units conducting financed emissions calculations for portfolio alignment shall develop specific implementation guidelines explaining the detailed methodology used, and that shall be consistent with this document.

3 GOVERNANCE OF THE GUIDELINES

3.1 Ownership of the guidelines

The Responsible Banking function is responsible for developing these guidelines.

They are approved in the ESG Working Group.

3.2 Financed emissions governance

The portfolio alignment financed emissions are approved when the decarbonization targets are approved at group level. Each decarbonization target has to be validated by the Responsible Banking governance bodies (The "Responsible Banking Forum" and the "Responsible Banking, Sustainability and Culture Committee") and approved the Banco Santander Board. For the local sectors, in addition to the required group governance, the targets (including its financed emissions) shall be approved by the local governance bodies.

Significant changes of Portfolio alignment financed emissions already approved: if a business unit responsible for managing a decarbonization target and the associated financed emissions want to make a significant change in the methodology or the scope of the calculations, it shall be validated by the global responsible banking function or in the ESG Working Group.

4 CHANGE CONTROL

ID	Owner	Maintenance	Validation	Approval Date
V.1	(Responsible Banking)	Etienne Butruille / Arnau Roselló	ESG Working Group	3 September 23

Date	Description
6 September 2023	First version